GG-M-108a
SEPTEMBER 25, 1956
SUPERSEDING
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February 23, 1956, and
Fed. Spec. GG-M-108
February 21, 1945

FEDERAL SPECIFICATION

# MANDRELS; DENTAL HANDPIECE

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.**—This specification covers dental handpiece mandrels for use with dental handpieces and contra-angles. The mandrels are used to mount the abrasive wheels and discs used for dental grinding and polishing procedures.

**1.2 Classification.**

1.2.1 *Types.*—Dental mandrels shall be of the three following types, as may be specified in the invitation for bids:

Type I.—Morgan-Maxfield.
Type II.—Plain screwhead.
Type III.—Compensating screwhead.

1.2.2 *Grades.*—Dental mandrels shall be of the two following grades, as may be specified in the invitation for bids:

Grade A.—Corrosion-resisting steel.
Grade B.—Brass, nickel plated.

1.2.3 *Classes.*—Dental mandrels shall be of the two following classes, as may be specified in the invitation for bids.

Class 1.—Angle.
Class 2.—Straight.

1.2.4 *Sizes.*—Dental mandrels shall be of one size except that type II shall be furnished in the two sizes shown in table II, as specified in the invitation for bids.

## 2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification.

*Federal Specifications:*

GG-I-526 — Instruments, Dental and Surgical; General Specifications.
LLL-B-631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL-B-636—Boxes, Fiber, Solid (for Domestic Shipment).
PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-585—Boxes; Wood, Wirebound.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-676—Boxes, Set-Up, Paperboard.

*Military Specifications:*

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL-L-10547 — Liners, Case, Waterproof.

*Military Standard:*

MIL-STD-129—Marking for Shipment and Storage.



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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

### 3.1 Material.

3.1.1 *Grade A.*—Shall conform to the requirements of Federal Specification, GG–I–526, class 2, composition H.

3.1.2 *Grade B.*—Shall conform to the requirements of Federal Specification, GG–I–526, class 4, composition N.

**3.2 Type designation.**—Mandrels shall conform in design and construction with the standard commercial type known to the dental industry by name or number designation as specified herein, and shall be suitable for use in all standard angle and straight dental handpieces, respectively. Class 1 mandrels shall also fit handpieces utilizing a slinger.

**3.3 Eccentricity.**—Over-all eccentricity of class 1 and class 2 mandrels shall be not more than 0.003 inch when tested as specified in 4.3.2.

**3.4 Shanks.**—Shanks for class 1 mandrels shall be 0.0919 to 0.0929 inch in diameter, and shanks for class 2 mandrels shall be 0.0919 to 0.0934 inch in diameter. Shanks of both classes shall conform to length measurements, plus or minus 0.01, as indicated in tables hereinafter specified. The shank end of class 1 mandrels shall be of the latch type and shall be cut away and notched as shown in figure 1. The shank end of class 2 mandrels shall be smoothly rounded.

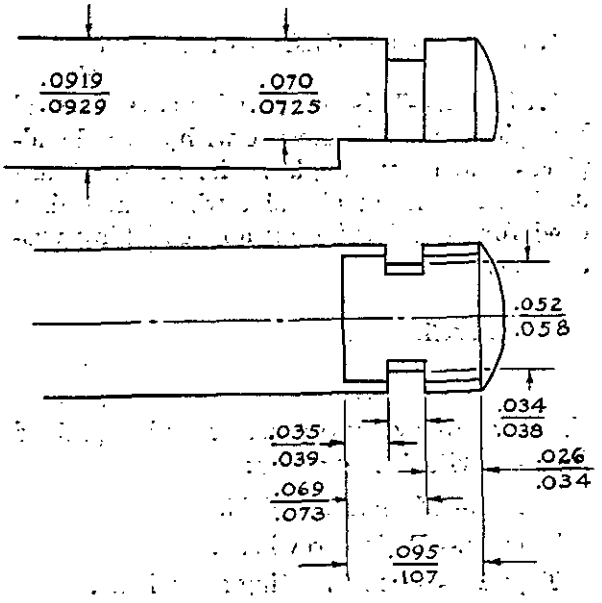



FIGURE 1.

**3.5 Identification marking.**—Unless otherwise specified, each mandrel shall be permanently marked in a suitable place with the name or trademark of the manufacturer or contractor.

### 3.6 Type I.

3.6.1 *Design.*—The Morgan-Maxfield mandrel shall be of the conventional type, equipped with a reinforced collar at one end and a removable pin-type head. Collar and shank shall be drilled for the reception of the pin spring attached to head. Collar shall be equipped with three short pointed pins spaced at equal intervals on the periphery of the



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collar, for securing disks in place. Head shall be cup-shaped and fit over pins on collar.

3.6.2 *Dimensions.*—Type I mandrels shall be furnished in the following classes, as specified in the invitation for bids, and shall conform to the requirements for dimensions given in table I.

TABLE I

| Class | Over-all length | Diameter of head | Diameter of pin |
|---|---|---|---|
| | *Inches* | *Inch* | *Inch* |
| 1 -------- | 1 | ¼ | 0.036 |
| 2 -------- | 1¾ | ¼ | 0.036 |

**3.7 Type II.**

3.7.1 *Design.*—Plain screw mandrels shall be of the conventional design with a collar at one end of shank. They shall be threaded for and be equipped with a screwhead of the same diameter as the collar in two sizes of head diameters.

3.7.2 *Dimensions.*—Type II mandrels shall be furnished in the following classes and sizes, as specified in the invitation for bids, and shall conform to the requirements for dimensions given in table II. The threaded portion of the screw head shall be of sufficient length to insure sufficient contact with female thread of mandrel when disks and wheels are attached.

TABLE II

| Class | Over-all length | Diameter of heads | | Diameter of screw | Threads per inch |
|---|---|---|---|---|---|
| | | Size 1 | Size 2 | | |
| | *Inches* | *Inch* | *Inch* | *Inch* | |
| 1----- | ¾ | 3/16 | ⅛ | 0.066 | 56 |
| 2----- | 1¾ | 3/16 | ⅛ | 0.066 | 56 |

**3.8 Type III.**

3.8.1 *Design.* — Compensating, screwhead mandrels shall be constructed with a steel pressure spring housed in a sliding ferrule mounted on the mandrel shank. The ferrule shall be free to rotate upon the shank without binding. The ferrule end of the mandrel shall be threaded for and equipped with a properly beveled screwhead of the same diameter as the ferrule. The spring shall be made of corrosion-resistant wire and shall maintain adequate pressure on the ferrule so as to cushion and lock disks or stones securely in place against the mounting screwhead.

3.8.2 *Dimensions.*—Type III mandrels shall be furnished in the following classes, as specified in the invitation for bids, and shall conform to the requirements for dimensions given in table III.

TABLE III

| Class | Length overall with spring compressed | Combined length of screwhead and ferrule | Diameter of head | Threads per inch |
|---|---|---|---|---|
| | *Inches* | *Inch* | *Inch* | |
| 1 -------- | 1 | ¼ | 3/16 | 56 |
| 2 -------- | 2 | ¼ | 3/16 | 56 |

**3.9 Workmanship.** — Shall be first class throughout and be free from any flaws such as tool marks, which detract from their appearance or which may impair their serviceability.

## 4. SAMPLING, INSPECTION AND TEST PROCEDURES

**4.1 Sampling.** — At least 1 container (6 mandrels) of each type, grade, and class shall be taken at random from every 500 packages or fraction thereof delivered for inspection and/or tests.



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**4.2 Inspection.**—Samples shall be carefully inspected for defects in material, workmanship, dimensions, finish, and other applicable requirements.

**4.3 Tests.**

4.3.1 *Dimensions.*—Measurements shall be made with a toolmaker's microscope, or other instrument of equal accuracy, when necessary.

4.3.2 *Eccentricity.* — The shank of the mandrel shall be rotated in a vee block by means of a round file. The eccentricity at the maximum diameter of the collar shall be measured by means of a dial gage. The total movement shown on the gage shall be the over-all eccentricity.

4.3.3 *Material.*—Tests shall be conducted in accordance with the applicable requirements of Federal Specification GG–I–526 for the grade and class specified.

## 5. PREPARATION FOR DELIVERY

(For definitions and levels, see 6.1).

**5.1 Packaging.**

5.1.1 *Unit of issue.*—Six (6) mandrels, as specified constitutes one unit.

5.1.2 *Quantities.*—Packaging and packing quantities shall be as specified in table IV.

TABLE IV.—*Packaging and packing quantities*

| Unit pack | Intermediate pack | Shipping container |
|---|---|---|
| *Unit* | *Units* | *Units* |
| 1 | 12 | 576 |

5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Six (6) mandrels (1 unit) shall be wrapped in neutral tissue and packaged in a carton of appropriate size and design.

5.1.3.2 *Intermediate package.*—The number of units specified in table IV shall be packaged in a carton of appropriate size constructed in accordance with Federal Specifications LLL–B–631, LLL–B–636, PPP–B–566, or PPP–B–676. Carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.*—Each unit shall be packaged in standard commercial containers of the size and kind commonly used for the purpose, which will afford the degree of protection required for shipment and use of the end product for its intended purpose within the continental limits of the United States of America.

**5.2 Packing.**

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—The number of units specified in table IV shall be overpacked in an exterior container designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; or PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, or MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

**5.3 Marking.**—Each unit package, each intermediate carton, and each shipping con-



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tainer shall be marked in accordance with Military Standard MIL–STD–129. Service color marking and symbol are not required on containers for level A packs, for the Medical Services of the Armed Forces.

## 6. NOTES

6.1 Federal Standard 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Purchasers should exercise any desired options offered herein and specify type, grade, class and size required. (See paragraphs 1.2.1, 1.2.2, 1.2.3, 1.2.4, 3.5, 3.6.2, 3.7.2, 3.8.2, 5.1 and 5.2).

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal Specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.5 **Transportation description.** — Transportation description applicable to this item is:

Mandrels, not otherwise indexed by name, iron or steel.

Carload minimum weight 30,000 pounds.

Truckload minimum weight 36,000 pounds.

6.6 This specification covers the following items appearing in the Armed Services Medical Stock List:

6520–540–7250 — Mandrel, Dental Handpiece, Angle Handpiece, Screwhead, 6s.

6520–540–8250 — Mandrel, Dental Handpiece, Straight Handpiece, Screwhead 6s.

**Patent notice.**—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

**MILITARY INTERESTS:**
**Army—M**
**Navy—Md**
**Air Force.**

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